Enhanced Multi-Instance Partial Label Learning via Average Gradient Outer Product

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Abstract

Multi-Instance Partial-Label Learning (MIPL) is a *dual* weakly supervised framework where each training bag is annotated with a candidate label set comprising a single ground-truth label and multiple false positives. The core challenge in MIPL is that the association between key instances and the ground-truth label is significantly obscured by these coarse and ambiguous supervision. Existing methods typically rely on model predictions to aggregate instance features via learned attention weights; however, this approach is susceptible to false positives, often causing key instances to be overlooked. To address this, we propose Average Gradient Outer Product based Multi-Instance Partial-Label Learning (AGOPMIPL). Specifically, we compute the Average Gradient Outer Product (AGOP) from the gradients of the bag-level classifier, integrating it into an attention module to amplify discriminative feature directions and facilitate key-instance identification. Furthermore, we introduce feature prototypes and a progressive disambiguation strategy to further suppress noisy candidates. Extensive experiments on four MIPL benchmarks and a real-world CRC-MIPL dataset demonstrate that AGOPMIPL consistently outperforms five state-of-the-art baselines, achieving a relative gain of up to 25.9% on CRC-MIPL-KMeansSeg.

1. Introduction

While supervised learning has achieved remarkable success, its reliance on exhaustive, high-quality annotations presents a severe bottleneck due to prohibitive labeling costs. This limitation has catalyzed the rise of weakly supervised learning (Zhou, 2018; Li et al., 2019; Nodet et al., 2021). Among these paradigms, Multi-Instance Learning (MIL) (Dietterich et al., 1997; Zhou et al., 2009; Zhang & Jin, 2020) has emerged as a prominent framework utilizing coarse bag-level labels. However, standard MIL often proves insufficient for complex real-world scenarios where supervision is not only coarse but also ambiguous. Consider image annotation via crowdsourcing: an image (bag) containing multiple objects often yields a candidate label set comprising both the ground truth and false positives. This scenario introduces a dual weak supervision: uncertainty regarding which instances are informative (instance space) and which label within the candidate set is correct (label space). To address this compounded challenge, Multi-Instance Partial-Label Learning (MIPL) (Tang et al., 2023) has been proposed, aiming to jointly disambiguate the label space and localize key instances under such weak supervision.

Existing MIPL methods generally fall into two categories, and both largely inherit their machinery from MIL and partial-label learning rather than addressing how the two ambiguities interact. Instance-level methods propagate the bag's candidate set down to each instance and disambiguate locally (Tang et al., 2024b), which relocates the partial-label problem onto the instances but forfeits the global bag context. Bag-level methods instead aggregate instances into a bag representation and disambiguate at the bag level (Tang et al., 2023; 2024a; Yang et al., 2024; 2025); crucially, the aggregation weights are read off inter-instance correlations or distributional priors, so the candidate labels supervise only the bag-level classifier and never tell the attention which feature directions distinguish the true label. Being label-agnostic, the attention simply follows the most salient or mutually correlated instances. This is precisely why noisy candidates are damaging: the false-positive labels corrupt the bag-level supervision, while the label-blind attention has no way to recover the true key instance, so attention drifts onto irrelevant ones and the key instance carrying the true label is overlooked. To validate this limitation, we evaluated four instance-attention-based MIPL methods on the FMNIST-MIPL dataset under varying noise ratios r. As shown in Figure 1, existing methods demonstrate limited efficacy in identifying key instances. Specifically,

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